# Unit XIV: Carboxylic Acids and Derivatives

## Video lecture notes: what she explained and wrote on the slides

CHEMISTRY (PROF. L. RANAY)

### WHERE THIS FITS

- **The last chapter of organic chemistry.** Four functional groups: **carboxylic acids, amides, esters, anhydrides**. For each: naming, physical properties, reactions.
- The carboxyl group is still a **carbonyl**, but with an **OH** on the carbonyl carbon (an aldehyde has an H there).
- Carboxylic acids are **really acids**, not neutral compounds. Reminder: amines are bases, carboxylic acids are organic acids.

### NAMING CARBOXYLIC ACIDS

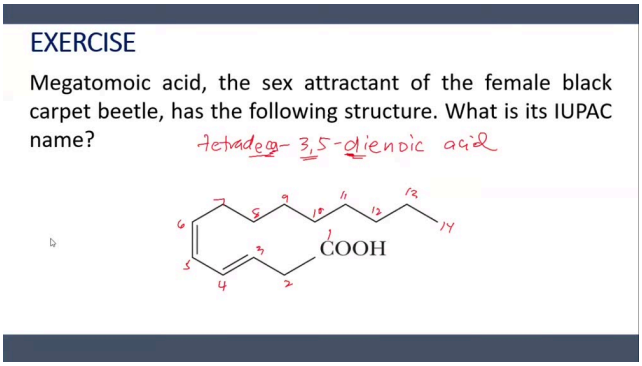
- Longest chain that contains the carboxyl group. **Drop the -e, add -oic acid**: methanoic acid, ethanoic, propanoic, butanoic, pentanoic, hexanoic...
- The COOH carbon is **always C1 (terminal)**: no locator, like an aldehyde.
- **Highest priority** functional group, so OH and $NH_2$ become **hydroxy-** and **amino-** prefixes.
  - **5-hydroxyhexanoic acid**; **4-aminobenzoic acid** (para-aminobenzoic acid, PABA).
  - Her examples: hexanoic acid (caproic), **3-methylbutanoic acid** (isovaleric).
- Numbering priority (repeated from the earlier chapters): **functional group → double bond → nearest branch → alphabetical → either side**.
- **Dicarboxylic acids:** add **-dioic acid** and **keep the -e**: **ethanedioic acid** (oxalic), **propanedioic** (malonic), **butanedioic** (succinic), pentanedioic (glutaric), hexanedioic (adipic). She said: you will meet these a lot in biochemistry, especially the **TCA / Krebs cycle**.
- Greek letters for common names: α, β, γ locate substituents. **4-aminobutanoic acid = γ-aminobutyric acid = GABA.**
- **Carboxylic acids are the most abundant organic compounds in nature**, and most of the common names we still use come from this family. **Fatty acids are carboxylic acids**, which is why the common names run up to C20.

### COMMON NAMES AND WHERE THEY COME FROM

- **Formic** (C1): Latin *formica*, ant. Found in the sting of ants and bees.
- **Acetic** (C2): Latin *acetum*, vinegar.
- **Propionic** (C3): Greek *propion*, first fat.
- **Butyric** (C4): Latin *butyrum*, butter.
- **Valeric** (C5): "strong" (Latin *valere*).
- **Caproic, caprylic, capric** (C6, C8, C10): Latin *caper*, goat.
- **Lauric** (C12): laurel; popular in coconut oil.
- Longer chains: tetradecanoic = C14 (myristic), hexadecanoic = C16 (palmitic), eicosanoic = C20.

### HER NAMING EXERCISE: MEGATOMOIC ACID

- The sex attractant of the female black carpet beetle: a 14-carbon chain from the COOH (C1) with **two C=C bonds** (she counted them at C3 and C5, "die-in").
- **Answer: tetradeca-3,5-dienoic acid** (3,5-tetradecadienoic acid).
  - The **-a- stays** in "tetradeca" because the next letter is a consonant (d of "diene"): just like hexanedioic or heptanedioic.
  - If it said "-en", the e would drop before a vowel; it doesn't here.

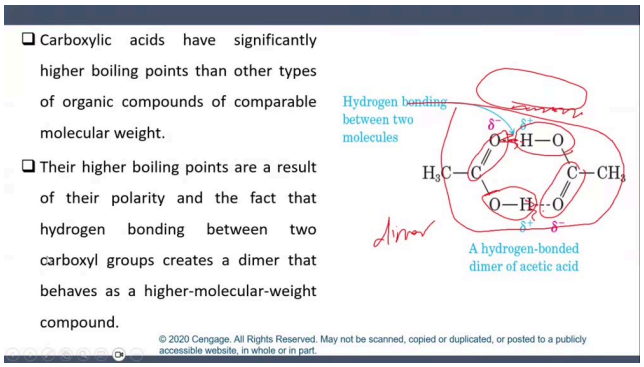



*Her answer on the slide: she numbered the chain 1 to 14 from the COOH and wrote tetradeca-3,5-dienoic acid*

**Note:** The recording makes it sound like "dioic", but she wrote **-dienoic** on the slide (two C=C bonds), not "-dioic" (that would mean two COOH groups).

## PHYSICAL PROPERTIES

- Very **polar** (two oxygens) and the O–H can **hydrogen-bond**. In addition they form **dimers**: two carboxylic acid molecules pair up by two hydrogen bonds (δ– O of one with δ+ H of the other), and the pair behaves like one bigger molecule. A dimer is "diatomic-like": two molecules acting as one. So there is an **extra strong intermolecular attraction**.
- Therefore the **highest boiling and melting points** of the organic families.
- **Boiling-point order from highest to lowest** (her list):
  - **Carboxylic acid > alcohol > amine > aldehyde/ketone > ether > thiol > hydrocarbon.**
- **Solubility:** H bonding with water. **Butanoic acid is infinitely soluble.** Her boundary: **C6 is slightly soluble, C7 and up are insoluble.** The more carbons, the more nonpolar the molecule, the less soluble.

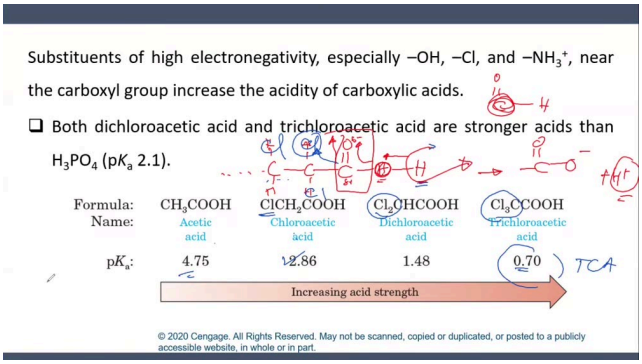



*Her drawing of the acetic acid dimer: two hydrogen bonds between two carboxyl groups, circled as one unit*

- **Exercise: arrange by increasing boiling point:** (b) the aldehyde < (c) the alcohol < (a) the carboxylic acid. Slide values: octanal 171 °C, 1-octanol 195 °C, octanoic acid 223 °C.

## ACIDITY

- Carboxylic acids are **weak acids** (organic acids are far weaker than HCl, $H_2SO_4$ or $HNO_3$), but real acids.
- **Among simple acids: methanoic > ethanoic > propanoic** (C1 > C2 > C3).
- **Why they are acids:** the O–H hydrogen leaves as $H^+$ because the **C=O pulls electrons through the carbon** and helps release it, leaving the carboxylate ion ($RCOO^-$).
  - In an **alcohol** the O pulls on the H, but alone it is not strong enough: **no $H^+$ is released** (except in phenols).
  - In an **aldehyde** the H is on carbon, which is not very electronegative: the weakest of the three.
  - **Carboxylic acid:** two pulls on the electron (the C=O and the O–H oxygen): enough to release $H^+$.
- **Any extra highly electronegative atom or group on the α-carbon (or any carbon near the COOH) pulls even more electron density, so the acid is STRONGER.** This is the **inductive effect**.
  - $CH_3COOH$ < $ClCH_2COOH$ < $Cl_2CHCOOH$ < **$Cl_3CCOOH$ (TCA, trichloroacetic acid, very acidic)**. Slide $pK_a$: 4.75, 2.86, 1.48, 0.70.
- The form in water depends on the **pH**: RCOOH at low pH (≤ 2), half and half at **pH = $pK_a$**, carboxylate $RCOO^-$ at pH ≥ 7.




*Her sketch for acidity: the electronegative atoms pull electrons, so H+ leaves more easily; TCA (pKa 0.70) circled as the strongest*

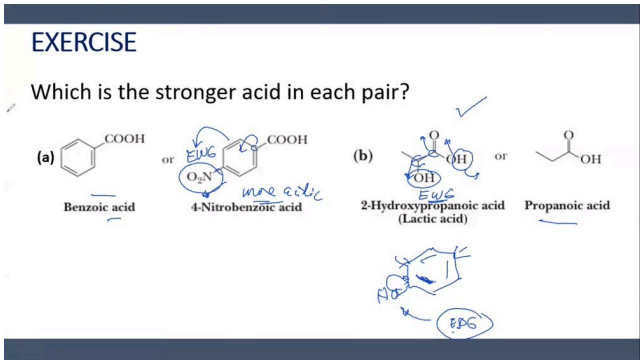
## HER TWO "STRONGER ACID" EXERCISES

### Benzoic acid or 4-nitrobenzoic acid?

**4-Nitrobenzoic acid.** $NO_2$ is an **electron-withdrawing group**, so it makes the acid stronger.

### 2-Hydroxypropanoic acid (lactic acid) or propanoic acid?

**Lactic acid.** The OH on the α-carbon is also electronegative and electron-withdrawing.



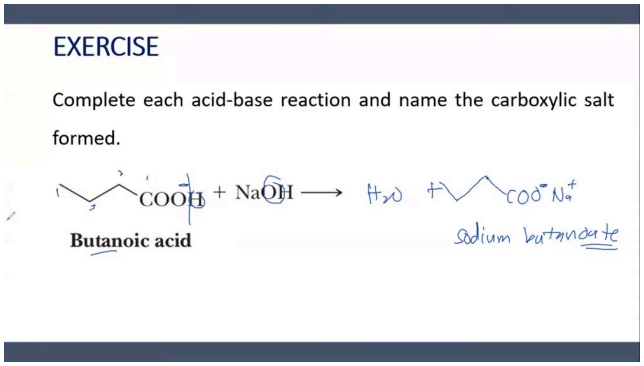

*Her marks on the pairs: -NO2 and the -OH of lactic acid labeled EWG (more acidic); a ring OH is an EDG*

- She stressed a **difference from the benzene chapter**: an OH **on a benzene ring donates** electrons (resonance with the ring, an EDG, ortho/para director). But here the OH is on a **saturated carbon**: no resonance, so it **withdraws** electrons.

## REACTIONS OF CARBOXYLIC ACIDS

### 1. WITH BASES: NEUTRALIZATION

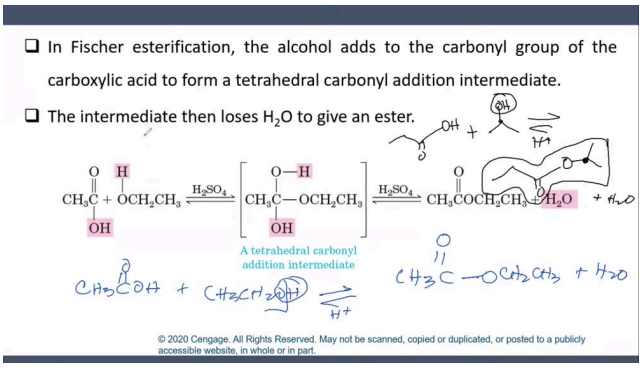
- Acid + base → **salt + water**. The $OH^-$ and $H^+$ make $H_2O$; the rest is a salt.
- **Benzoic acid + NaOH → sodium benzoate + $H_2O$.** She noted sodium benzoate is a **food preservative**.
- With ammonia (a base): **ammonium benzoate**.
- **Naming the salt: -oic acid → -oate.** Exercise: **butanoic acid + NaOH → sodium butanoate + $H_2O$.** "When the -oic acid becomes an anion, the name becomes -oate: no longer an acid because the H is gone."
- (Slide: with $NaHCO_3$ you get the sodium salt + $H_2O$ + **$CO_2$ gas**.)




*Her answer to the exercise: butanoic acid + NaOH gives sodium butanoate + H2O*

### 2. WITH ALCOHOLS: ESTERIFICATION (FISCHER ESTERIFICATION)

- Carboxylic acid + alcohol (acid catalyst, conc. $H_2SO_4$) → **ester + water**. This is how esters are made.
- **Which atoms leave:** the **OH from the carboxylic acid** and the **H from the alcohol**. They form the water. **Not the other way around.**
- The **alcohol's oxygen ends up between the carbonyl carbon and the alkyl group** (the ester oxygen).
- She compared it with hemiacetal formation: the carbonyl carbon is δ+ and the alcohol's O attaches to it. **But here the π bond does not break**: it is a **substitution** (an OH is replaced by OR), not an addition.
- **Reversible**; it can be driven either way. If the water is not removed it stays an equilibrium (reverse = ester hydrolysis).
- Example on the board: **ethanoic acid + ethanol ($H^+$) → ethyl ethanoate + $H_2O$**.
  - Second example (spoken): an alcohol whose **OH is on the second carbon** (like propan-2-ol). The OH of the acid is the one removed, so **the alcohol's O stays on that middle carbon**: in the product the ester oxygen is attached to the **middle carbon** of the alkyl group (that is why her board name for it is **isopropyl ethanoate**). Water is the by-product.




*Her Fischer esterification board work: CH3COOH + CH3CH2OH with H+ gives CH3COOCH2CH3 + H2O; the leaving OH and H circled*

- **Slide exercise:** a **diacid + excess ethanol ($H^+$)** gives the **diester** (both COOH groups become COOEt).

### 3. DECARBOXYLATION

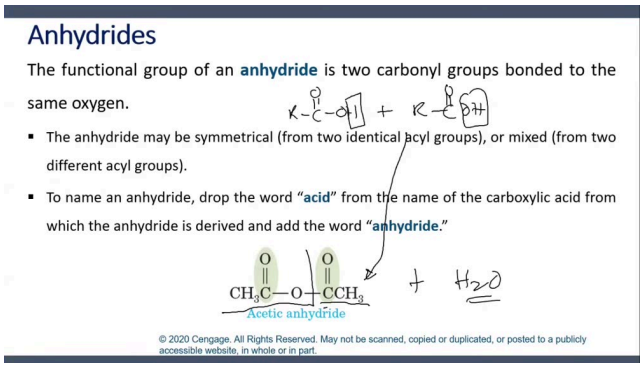
- **Loss of $CO_2$** from the carboxyl group. Without enzymes it needs **very high temperature**; simply

melting the acid is not enough.

- In the body, **enzymes** do it at body temperature. Example: **3-oxobutanoic acid → acetone + $CO_2$** (a β-keto acid loses $CO_2$ easily).
- The $CO_2$ **we exhale** comes partly from decarboxylation in the **Krebs cycle**.

## CARBOXYLIC ACID DERIVATIVES

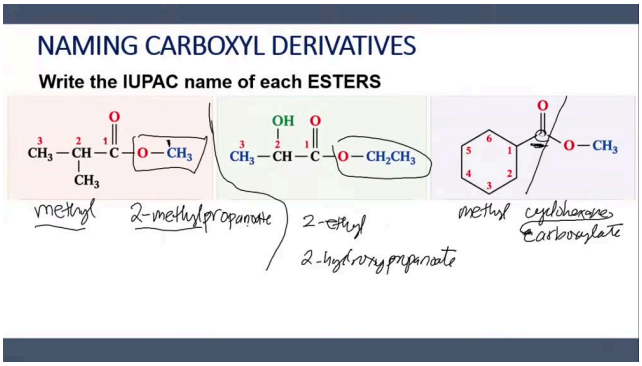
- **Anhydride, ester, amide.** (Her mnemonic: **"see anhydride, see ester, see amide"**: C-an, C-es, C-am.)
- **Anhydride:** from **two carboxylic acids** by **removal of water** ("anhydrous" = without water). The functional group is **an oxygen between two carbonyl carbons**.
- **Ester: carbonyl + alkoxy (–OR)**, made from a carboxylic acid and an alcohol (Fischer esterification).
- **Amide: carbonyl + nitrogen**.
- **Cyclic amide:** a **β-lactam**, very important in biochemistry (penicillin-type antibiotics on the slide).
- All three are called **derivatives** because **hydrolysis gives back the carboxylic acid** (see below).




*Her anhydride sketch: two carboxylic acids, OH and H leave as H2O, one O stays between the two carbonyls*

## NAMING ESTERS

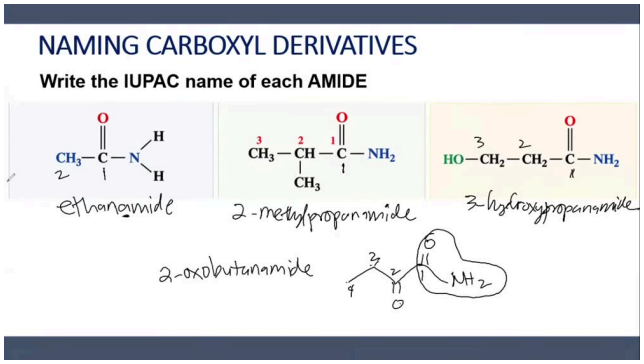
- An ester name has **two parts, written as two words (with a space, no dash)**:
  - The **alcohol (O-attached) part** gets **-yl**.
  - The **C=O (acid) part** gets **-oate**.
  - **Ethyl ethanoate** ($CH_3COOCH_2CH_3$). **Isopropyl ethanoate**. **Methyl butanoate**.
- Number each part **separately, each starting from its own end**: the acid part starts from the C=O carbon; the alcohol part from the carbon bonded to O.
  - **Methyl 2-methylpropanoate** (the 2-methyl is on the acid part); **ethyl 2-hydroxypropanoate** ("it should be a space, not a dash").
- **Ring on the carbonyl carbon (C outside the ring):** aldehyde → **carbaldehyde**; ester → **carboxylate**: **methyl cyclohexanecarboxylate** (one word for the acid part).
- **Esters smell sweet** (fruits like apple and pineapple, perfumes). **Thiols smell bad.**



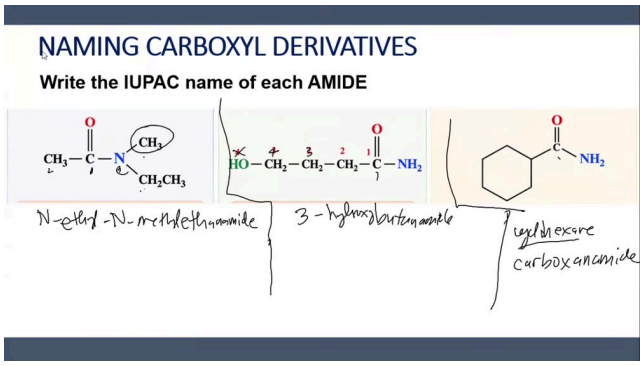

*Her ester names on the board: ethyl ethanoate, isopropyl ethanoate, methyl butanoate, methyl 2-methylpropanoate, ethyl 2-hydroxypropanoate, methyl cyclohexanecarboxylate*

## NAMING AMIDES

- The C=O carbon is C1: **drop -oic acid, add -amide**: ethanamide, **2-methylpropanamide**, **3-hydroxypropanamide** ($HO–CH_2–CH_2–CONH_2$ on her slide).
- **Do not locate the functional group** in esters and amides: the carbonyl is always C1. **From the carboxylic acid down to its derivatives the carbonyl is always number one; the only carbonyl you must locate is a ketone.**
- A ketone C=O inside an amide chain becomes **oxo-**: **2-oxobutanamide** (the amide C=O is the main group).
- **Groups on the nitrogen use N- (capital N)**, alphabetical: **N-ethyl-N-methylethanamide**. She said amide naming is **easier than amine naming**: you never have to locate the nitrogen.
- **Ring: cyclohexanecarboxamide** (aldehyde: carbaldehyde; ester: carboxylate; amide: carboxamide). The C=O carbon is outside the ring.




*Her amide names: ethanamide, 2-methylpropanamide, 3-hydroxypropanamide, 2-oxobutanamide*

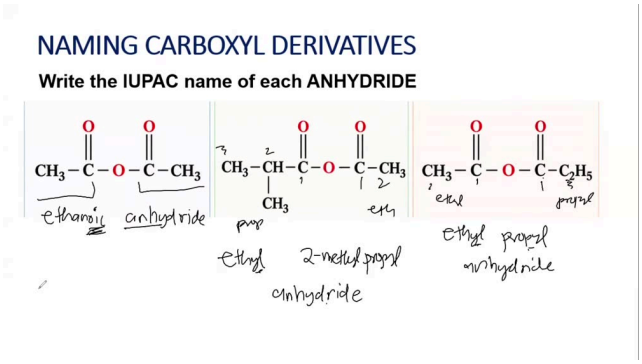



*N-ethyl-N-methylethanamide and cyclohexanecarboxamide*

## NAMING ANHYDRIDES

- **Symmetrical** (same acid on both sides): name the acid, replace **"acid" with "anhydride"**: **ethanoic anhydride**, propanoic anhydride.
- **Unsymmetrical (two different acids):** she used the **-yl names** of the two groups, **alphabetical, then "anhydride"**: **ethyl propyl anhydride** (as she wrote it); a branched one: "ethyl (2-methylpropyl) anhydride".

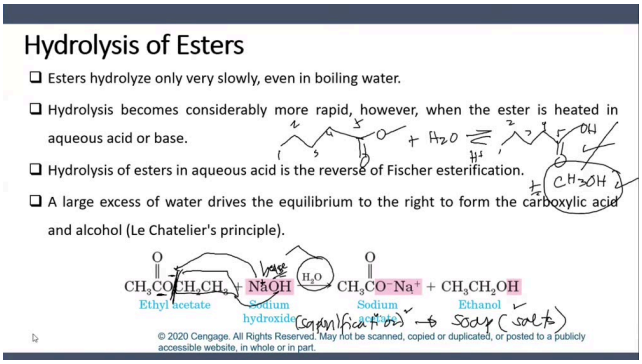
**Note:** Standard IUPAC would call the unsymmetrical ones "ethanoic propanoic anhydride". The question bank only quizzes the **symmetrical** rule (-oic anhydride) and the idea that two different acids give a mixed anhydride.




*Her anhydride names: ethanoic anhydride, ethyl 2-methylpropyl anhydride, ethyl propyl anhydride*

## HYDROLYSIS (REACTION WITH WATER; HYDRO = WATER, LYSIS = BREAKDOWN)

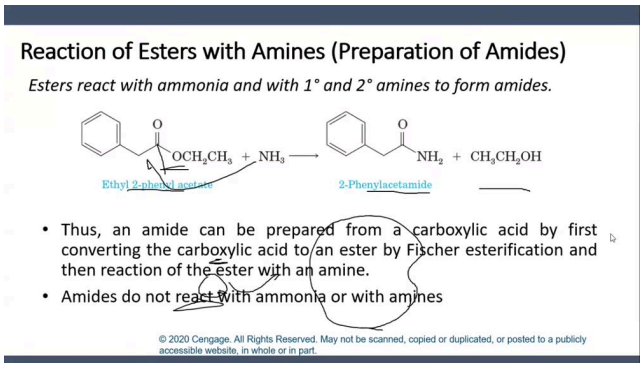
- **Anhydride + water → two carboxylic acids** (very fast for small ones).
  - She used this to explain: **acetic anhydride is NOT "soluble" in water**: it reacts instead of dissolving and becomes two acetic acids.
- **Ester hydrolysis:**
  - **In base (NaOH): carboxylate salt + alcohol = saponification**, the soap reaction. **Soaps are salts of carboxylic acids.** ($Na^+$ is a cation, so it pairs with the carboxylate anion.)
  - **In water with an acid catalyst: carboxylic acid + alcohol.** This is the **reverse of Fischer esterification** (esterification is reversible).
- **Amide hydrolysis:** water + acid and heat: **carboxylic acid + ammonium salt** ($NH_3 + HCl \rightarrow NH_4^+Cl^-$; the nitrogen part reacts with the acid's anion to form a salt). Base hydrolysis also exists but acid is the common one.
- Because **esters and amides both give a carboxylic acid on hydrolysis**, they are "derivatives".




*Her ester hydrolysis notes: with acid and water, the acid + CH3OH; with NaOH (saponification) the soap, a salt*

## OTHER REACTIONS OF DERIVATIVES

- **Anhydride + ammonia or amine → amide + a salt (two moles of amine needed):** one forms the amide, the second neutralizes the carboxylic acid by-product (acetamide + ammonium acetate). She wrote "Salt" next to it.
- **Anhydride + alcohol → ester + carboxylic acid.** Anhydrides break easily, so they are handy to make esters.
  - **Aspirin** = salicylic acid (its OH) + acetic anhydride → acetylsalicylic acid + acetic acid.
  - Story she told: the first painkiller came from **willow bark** (salicin → salicylic acid). Salicylic acid is **corrosive to the stomach**, so chemists "acetylated" the OH: **less acidic = aspirin**.
- **Ester + amine (or ammonia) → amide + alcohol.** (On the slide she circled: an amide is prepared from a carboxylic acid by first making the ester, then reacting it with an amine; **amides do NOT react with ammonia or amines**.)




*Her circles on the slide: ester + amine gives an amide; amides do not react with ammonia or amines*

## POLYMERS (STEP-GROWTH)

- Polymers from two monomers, **each with two functional groups**. A new bond is made at each step.
- **Polyamides:** a diacid + a diamine; the link is an **amide**. **Proteins are polyamides**: the bond between amino acids is the **peptide bond (an amide)**. **Nylon-66** (first purely synthetic fiber, from two six-carbon monomers) and **Kevlar** (bulletproof vests; aromatic diacid + aromatic diamine).
- **Polyesters:** a diacid + a diol. Fabrics: **Dacron**; **Mylar** (balloons).
- **Polycarbonates: Lexan**, very hard plastic; from the disodium salt of **bisphenol A** and **phosgene** (an acyl halide). Used for **eyeglass lenses** (no longer glass).

## WHAT TO REVIEW

- Naming (acid, dioic, ester as two words, amide with N-, anhydride), boiling-point order, why acids are acidic and how EWGs raise acidity, neutralization, Fischer esterification (which atoms leave), decarboxylation, hydrolysis of all three derivatives, saponification, aspirin, the three polymer types.

END OF LESSON